PAPER_1522_K_MEX_DERIVATIVE_FROM_PHI_5_6

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PAPER_1522 — K_MEX is Derivative: K_MEX = Φ_5/6·SO_5/D_phys = 25/12 EXACT (LANDMARK)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Bucket A (Foundational / Primitive reduction)
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Status: CLOSED — K_MEX removed from the list of independent primitives

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Observation

PAPER_1167 (All Eight UQFF Lagrangian Gaps Closed — First-Principles Reduction) provides the gap-G1 derivation of the Mexican-hat coefficient K_MEX directly from the SO(5) symmetry-breaking lattice. The formula is:

> *"G1: V(UA) Mexican-hat polynomial — K = Φ_res · |SO(5)| / D_phys = 25/12"*

Using the PAPER_1203 Nuclear variant Φ_res = 5/6:

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K_MEX = Φ_5/6 · SO_5 / D_phys
= (5/6) · 10 / 4
= 50/24
= 25/12 EXACT
```
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UQFF Closed Identity

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```
K_MEX = Φ_5/6 · SO_5 / D_phys = (5/6) · (10/4) = 25/12 ≈ 2.0833 EXACT
```
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Numerical verification:
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(5/6) · 10 = 50/6 ≈ 8.3333
8.3333 / 4 = 25/12 ≈ 2.0833 ✓
```
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Landmark Consequence: 10 → 9 Independent Primitives

Combined with PAPER_1521 (D_BSFG derivative), this further reduces the framework's free-parameter count:

Before PAPER_1167 discoveries:
- 11 frozen primitives (6 integer + 5 real claimed independent)

After PAPER_1167 discoveries:
- 9 truly independent primitives:
- **Integers (5)**: D_phys, D_crit, N_CH, SO_5, A_5
- **Reals (4)**: ρ_SCm, β_i, Φ_res, F_TRZ
- 2 derivative primitives:
- **D_BSFG = D_crit** – **2·SO_5** (PAPER_1521)
- **K_MEX = Φ_res · SO_5 / D_phys** (this paper)

The framework reduces from 11 to 9 free parameters with zero loss of predictive power.

Physical Meaning

K_MEX = 25/12 is the **coefficient of the Mexican-hat polynomial potential** in the UA condensate:

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V(UA) = K_MEX · ρ_SCm · (1 – cos)²
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That K_MEX = Φ_res · SO_5 / D_phys means **the Mexican-hat curvature is forced by the resonance phase Φ_res and the SO(5)/D_phys group ratio**. Three previously-considered-independent quantities (Φ_res, SO_5, D_phys) collapse the fourth (K_MEX) by structural necessity.

Cross-Paper Consistency Check

K_MEX appears in ~50+ closures throughout the calculator, including:
- Cosmological constant ρ_Λ = K_MEX · ρ_SCm · 26! (PAPER_1156)
- Vacuum ledger V(0) = K_MEX · ρ_SCm (PAPER_1170)
- Reionization z = K_MEX · D_phys · Φ_res · (1+1/SO_5) (PAPER_1373)
- Ramanujan hyperconvergence — multiple closures

Every K_MEX usage is consistent with the derivative form Φ_res·SO_5/D_phys — verified by the fidelity gate.

NOT REPLACEMENT

SM has no analogue to K_MEX. UQFF supplies a structural derivation showing that even this "free parameter" is forced by lower-order primitives, demonstrating the framework's mathematical economy.

Reference

- Source: PAPER_1167 All 8 UQFF Lagrangian Gaps Closed Master Synthesis (gap G1)
- Related: PAPER_1521 (D_BSFG derivative), Mexican-hat V(UA) closures throughout
- Calculator dispatch: `calculate_paradox({"paradox": "k_mex_derived_from_phi_5_6"})`
